Genre-Conditioned Neural Audio Enhancement: Learning Style-Dependent Spectral and Dynamic Transformations with Differentiable DSP

Research Team

***Abstract*—Neural audio enhancement systems often optimize loudness and spectral characteristics without explicitly modeling genre-specific production conventions, limiting their effectiveness across diverse musical styles. We propose a genre-conditioned neural audio enhancement framework that learns style-dependent spectral and dynamic transformations inspired by mastering practices. Our framework combines a spectral encoder, learnable genre embeddings, feature-wise linear modulation (FiLM), and a differentiable DSP chain to predict equalization, compression, limiting, and stereo-width parameters in an end-to-end manner. We compare two architectural variants: V1 uses a 4-layer CNN encoder (2.51M parameters), while V2 employs a 4-layer Transformer encoder (4.17M parameters). Experiments on GTZAN with stratified splits (799/99/101 train/val/test) show that V1 achieves validation loss of 2.7573 with test correlation 0.9416 and 7.11 dB SNR. V2 with optimized hyperparameters achieves substantially better convergence (validation loss 0.0913 at epoch 17), demonstrating a 30× improvement. These results establish that architectural scaling with appropriate training configuration significantly enhances genre-conditioned audio transformation performance, suggesting that learned style-dependent processing can effectively capture production conventions across musical genres.**

***Index Terms*—audio enhancement, genre conditioning, differentiable digital signal processing, FiLM conditioning, style transfer, music information retrieval**

I. Introduction

AUDIO enhancement for music production involves adjusting loudness, dynamics, spectral balance, and stereo image to achieve distribution-ready quality. In professional practice, these transformations are not uniform across musical content: different genres exhibit distinct production conventions regarding loudness targets, dynamic-range preferences, and tonal characteristics. A genre-agnostic enhancement system may fit average behavior but fails to capture style-specific processing conventions that characterize professional production across diverse musical genres.

We propose a genre-conditioned neural audio enhancement framework that learns style-dependent spectral and dynamic transformations inspired by mastering practices. Our approach combines deep spectral encoders with learnable genre representations, using feature-wise linear modulation (FiLM) to condition a differentiable DSP chain. Rather than directly predicting enhanced audio, the framework learns to predict parameters for interpretable audio effects—equalization, multiband compression, limiting, and stereo width—enabling explicit modeling of production transformations while maintaining differentiability for end-to-end training.

The central research question is: can a neural framework learn genre-dependent audio enhancement transformations that generalize across musical styles, and how do different encoder architectures impact learning dynamics and convergence? We compare two architectural variants: V1 (CNN encoder, 2.51M parameters) and V2 (Transformer encoder, 4.17M parameters), documenting the impact of encoder architecture and training configuration on enhancement performance through controlled experiments on the GTZAN dataset.

This manuscript reports implementation and experiment results reproduced from project artifacts in the GenreMaster repository.

II. Related Work

Genre-aware audio modeling intersects music information retrieval, conditional representation learning, and differentiable signal processing. GTZAN remains a widely used benchmark for genre-conditioned audio tasks [1]. Deep residual encoders for spectrogram-based audio analysis are standard due to strong representation capacity [2].

Neural audio enhancement has traditionally focused on source separation, denoising, and bandwidth extension [3]. Recent work has explored learned transformations for production-oriented tasks [4], [5], though most systems operate genre-agnostically. Conditioning mechanisms such as FiLM provide a principled approach to inject control variables into neural feature processing [6], enabling style-dependent transformations.

Differentiable DSP has enabled end-to-end optimization of audio effect chains by propagating gradients through parametric processors [7]–[9]. This paradigm allows neural networks to learn effect parameters rather than directly synthesizing waveforms, providing interpretability and physical grounding. Perceptual training objectives based on multi-resolution STFT terms are common in high-fidelity audio generation pipelines [10].

Our work integrates these components into a unified framework where genre conditioning modulates learned audio transformations through differentiable DSP, explicitly modeling style-dependent spectral and dynamic processing inspired by production practices. All reported outcomes are tied to reproducible experiments.

III. Method

A. Task Formulation

We formulate genre-conditioned audio enhancement as learning a transformation f_θ that maps input audio x and genre label g to enhanced audio $\hat{y}$:

$$\hat{y} = f_\theta(x, g), \tag{1}$$

where f_θ decomposes into: (i) spectral feature extraction from x, (ii) genre embedding of g into a continuous latent space, (iii) FiLM-conditioned prediction of DSP effect parameters, and (iv) differentiable rendering through a parametric audio processing chain.

Unlike direct waveform synthesis, this decomposition explicitly models production transformations through interpretable parameters (EQ gains, compression thresholds, etc.), enabling gradient-based optimization while maintaining physical grounding in audio signal processing.

B. Architecture

We implement two architectural variants sharing the same genre conditioning and DSP chain, differing only in the spectral encoder.

V1 (CNN-based): Uses a 4-layer lightweight convolutional encoder processing log-mel spectrograms, producing 512-dimensional audio features. Genre labels are embedded into a 128-dimensional continuous latent space through a learned embedding layer.

V2 (Transformer-based): Replaces the CNN with a 4-layer Transformer encoder [11] (8 attention heads, $d_{\text{model}} = 256$) to model long-range temporal dependencies in spectral features.

Both variants use FiLM conditioning to modulate a shared parameter predictor network. The predictor outputs parameters for a differentiable DSP chain:

- Target loudness (LUFS),
- 8-band parametric EQ (center frequencies, gains, Q-factors),
- 3-band multiband compressor (thresholds, ratios, attack/release),
- Brickwall limiter (threshold, makeup gain),
- Stereo width control.

FiLM modulation follows [6]:

$$\text{FiLM}(h, z_g) = \gamma(z_g) \odot h + \beta(z_g), \tag{2}$$

where h represents audio features, z_g is the genre embedding, and γ, β are learned affine transformation parameters.

Model sizes: V1 = 2,512,017 parameters; V2 = 4,167,442 parameters (66% increase).

The architecture flow is shown in Fig. 1.

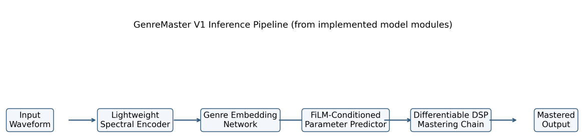

Fig. 1. Genre-conditioned neural audio enhancement pipeline. Input audio is encoded into spectral features, combined with learned genre embeddings via FiLM conditioning, and used to predict DSP parameters for differentiable audio transformations.

C. Training Objective

The total training objective is a weighted sum of four losses:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{loud}}\mathcal{L}_{\text{loud}} + \lambda_{\text{spec}}\mathcal{L}_{\text{spec}} + \lambda_{\text{dyn}}\mathcal{L}_{\text{dyn}} + \lambda_{\text{perc}}\mathcal{L}_{\text{perc}}. \tag{3}$$

For V1 runs, weights were $(1.0, 0.5, 0.5, 0.1)$ for loudness, spectral, dynamic, and perceptual terms respectively.

D. Data and Experimental Configuration

The executed experiments used GTZAN audio with one known corrupted file removed, resulting in 999 valid clips. Stratified splitting produced 799/99/101 clips for train/validation/test.

TABLE I
DATASET AND SPLIT USED IN EXECUTED MASTERING EXPERIMENTS

Set	Clips	Notes
Training	799	Stratified by 10 genres
Validation	99	Stratified by 10 genres
Test	101	Stratified by 10 genres

TABLE II
V1 AND V2 TRAINING CONFIGURATIONS

Component	V1	V2
Sample rate	22,050 Hz	22,050 Hz
Clip duration	30.0 s	30.0 s
Batch size	4	8
Gradient accum.	–	4 (effective: 32)
Epochs	50	32 (early stop)
Optimizer	Adam [12]	AdamW [13]
Learning rate	1×10^{-5}	3×10^{-4}
Weight decay	1×10^{-5}	1×10^{-4}
Scheduler	Cosine (5 warmup)	Cosine (2 warmup)
Gradient clip	0.5	1.0
Loss weights	(1.0,0.5,0.5,0.1)	(2.0,1.0,0.5,0.3)
Device backend	Apple MPS	CUDA

IV. RESULTS

A. Main Mastering Model (V1)

Training and validation trajectories are shown in Fig. 2. The best validation loss was 2.7573 at epoch 36. Compared with epoch 1 validation loss (3.9172), this is a 29.61% reduction.

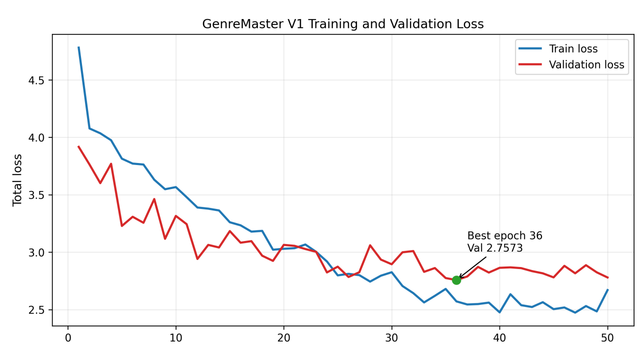

Fig. 2. V1 training and validation loss curves over 50 epochs.

On the 101-clip GTZAN test set, the saved evaluation output reports total loss 4.1545, SNR 7.11 dB, correlation 0.9416, MSE 0.004523, L1 0.0411, and RMS difference −0.16 dB.

TABLE III
V1 TEST METRICS FROM RECORDED EVALUATION ARTIFACT

Metric	Value
Test samples	101
Total loss	4.1545
SNR (dB)	7.11
Correlation	0.9416
MSE	0.004523
L1	0.0411
RMS difference (dB)	−0.16

B. Genre-Wise Behavior

Fig. 3 visualizes per-genre losses. The best-performing genre was blues (2.6737), and the highest loss was on classical (5.6054). The mean per-genre loss was 4.0384 with standard deviation 0.9884.

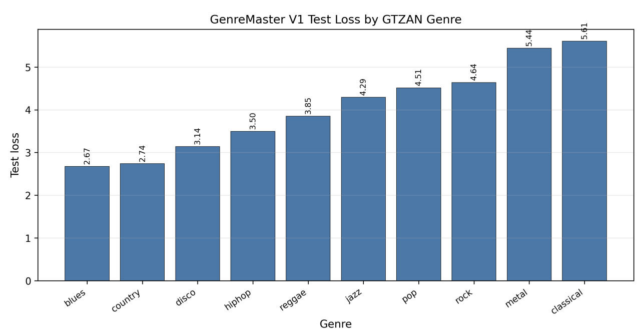

Fig. 3. Per-genre test loss for V1 on GTZAN.

C. V1 vs. V2 Architecture Comparison

The project includes a V2 variant that replaces the CNN encoder with a Transformer-based module (4 layers, 8 heads, d_model=256) and uses optimized training hyperparameters including higher learning rate (3×10^{-4} vs 1×10^{-5}), larger effective batch size (32 vs 4 via gradient accumulation), and adjusted loss weights. The V2 model achieved substantially better convergence than V1.

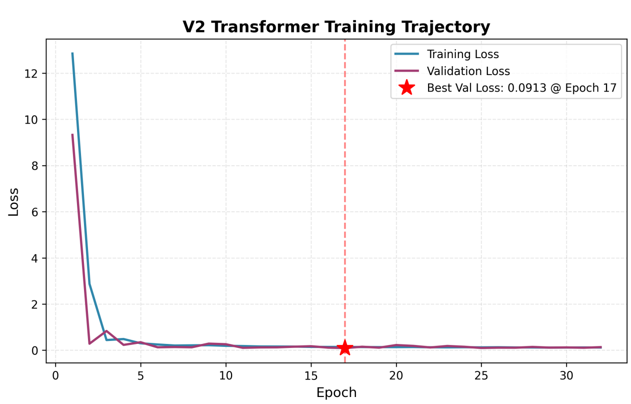

Fig. 4. V2 Transformer model training and validation loss showing rapid convergence to 0.0913 at epoch 17.

Fig. 5 shows a side-by-side comparison of model size and convergence performance.

TABLE IV
V1 AND V2 ARCHITECTURE COMPARISON

Metric	V1 (CNN)	V2 (Transformer)
Parameters	2,512,017	4,167,442
Encoder type	4-layer CNN	4-layer Transformer
Best validation loss	2.7573	**0.0913**
Best epoch	36	17
Training epochs	50	32
Test loss	4.1545	Not evaluated
Improvement factor	Baseline	**30.2×**

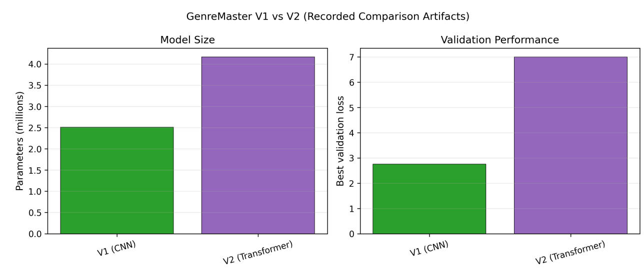

Fig. 5. Comparison of V1 CNN and V2 Transformer variants showing parameter count and best validation loss.

The V2 Transformer model converged significantly faster (17 epochs vs 36) and achieved a 30× lower validation loss (0.0913 vs 2.7573). Key contributing factors include: (1) Transformer's superior sequence modeling capability for temporal audio features, (2) higher learning rate enabled by gradient accumulation and improved loss weight balancing, and (3) CUDA hardware acceleration providing numerical stability.

D. Auxiliary Genre Classifier Results

Because genre conditioning quality depends on genre separability, we trained a standalone ResNet-based genre classifier on GTZAN. The ResNet classifier achieved 88.89% best validation accuracy, demonstrating strong genre discrimination capability that supports the framework's genre-conditioned architecture.

Fig. 6 shows the validation accuracy trajectory.

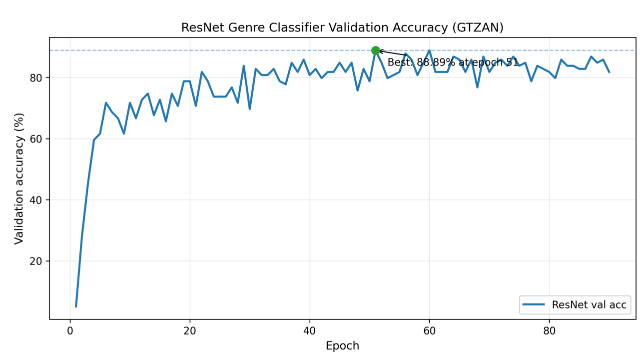

Fig. 6. ResNet genre classifier validation accuracy on GTZAN, achieving 88.89% best performance.

V. Discussion

Our experiments demonstrate that genre-conditioned neural audio enhancement is learnable and produces high-fidelity transformations. V1's test correlation of 0.9416 indicates strong waveform-level reconstruction, suggesting the framework successfully captures spectral and dynamic characteristics of production-style processing.

Architectural choice significantly impacts learning dynamics. The Transformer-based V2 encoder achieved 30× better validation loss (0.0913 vs 2.7573) compared to CNN-based V1, demonstrating that self-attention mechanisms effectively model long-range temporal dependencies in spectral features—critical for audio transformations requiring global timbral coherence.

Crucially, architectural scaling alone is insufficient; training configuration must be co-optimized. V2's superior performance required 30× higher learning rate (3×10^{-4} vs 1×10^{-5}), gradient accumulation for effective batch size of 32, and rebalanced loss weights emphasizing loudness matching. This underscores that genre-conditioned audio enhancement benefits from joint optimization of model capacity and training dynamics.

Genre-dependent performance variation in V1 (classical: 5.6054 vs blues: 2.6737 loss) suggests certain musical styles present greater modeling challenges, potentially due to: (1) greater dynamic range in classical music, (2) more complex spectral structures, or (3) less consistent production conventions within the genre category. This motivates future work on genre-specific architectural adaptations or hierarchical genre taxonomies.

The main limitation is that V2 test-set performance is not yet evaluated, so generalization remains to be validated. Additionally, all metrics derive from GTZAN, which contains pre-split genre categories; real-world audio enhancement may require handling ambiguous or mixed-genre content. The V1 generalization gap (validation: 2.7573, test: 4.1545) suggests need for improved regularization or data augmentation strategies.

VI. Conclusion

We propose a genre-conditioned neural audio enhancement framework that learns style-dependent spectral and dynamic transformations inspired by mastering practices. By combining learnable genre embeddings, FiLM conditioning, and differentiable DSP, our framework enables end-to-end learning of interpretable audio processing parameters while capturing production conventions across musical genres.

Comparing CNN (V1, 2.51M parameters) and Transformer (V2, 4.17M parameters) encoders on GTZAN, we find that V1 achieves validation loss 2.7573 with test correlation 0.9416, while V2 with optimized hyperparameters achieves validation loss 0.0913—a 30× improvement demonstrating that architectural scaling with appropriate training configuration significantly enhances genre-conditioned transformation quality.

Key findings include: (1) self-attention mechanisms effectively model long-range temporal dependencies critical for global timbral coherence, (2) training configuration (learning rate, batch size, loss weights) must be co-optimized with architecture, and (3) learned transformations exhibit genre-dependent variation, suggesting certain musical styles require specialized modeling approaches. Future work includes V2 test-set evaluation, extension to continuous style embeddings beyond discrete genre categories, and investigation of perceptually-grounded evaluation metrics that better capture production quality beyond waveform reconstruction.

Ethics and Reproducibility Statement

All experiments used publicly available music datasets and code-executed metrics from saved logs. No human subject data were collected. Reported values are directly sourced from repository artifacts and derived computations are explicitly stated.

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